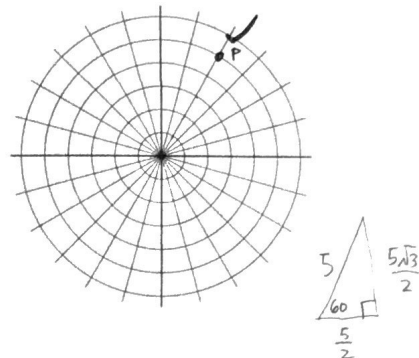
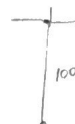


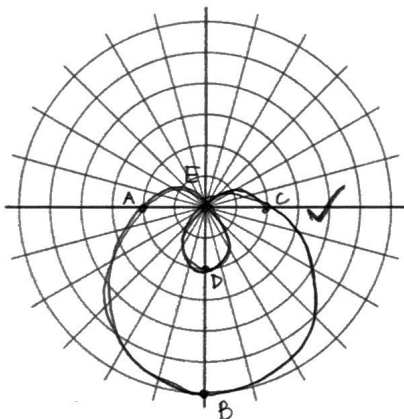
1. a) On the axes below, sketch point P that has polar coordinates: $(5, \frac{-5\pi}{3})$ [1pt]
b) Rename point P with a positive angle and negative radius $(-5, \frac{4\pi}{3})$ [1pt]
c) Convert point P into rectangular coordinates (simplified) $(\frac{5}{2}, \frac{5\sqrt{3}}{2})$ [1pt]



2. Convert the rectangular point $(0, -100)$ into simplified polar coordinates $(100, \frac{3\pi}{2})$ [2pts]



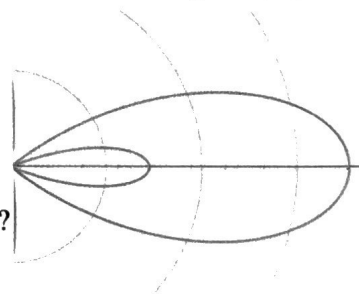
3. Accurately graph polar curve $r = -2 - 4\sin\theta$ by making a table of values. **Label** at least four different points on your graph using letters A, B, C, and D and show their polar coordinates in the table below. You may use the additional columns in the table to determine more points (this is optional) [6pts]



Name of point	A	B	C	D	E	
θ	0 ✓	$\frac{\pi}{2}$ ✓	π ✓	$\frac{3\pi}{2}$ ✓	0	
r	-2 ✓	-6 ✓	-2 ✓	2 ✓	0	

4. The graph of $r = 2 + 6\cos 3\theta$ has an interesting shape. The diagram below (not drawn to scale) shows a part of the curve, including a “petal inside a petal” along the angle $\theta = 0$.

$$\frac{6}{4} = \frac{3}{2}$$



- a) How long is the longer petal along $\theta = 0$? 8 [1pt]

- b) How long is the shorter petal along $\theta = 0$? 4 [1pt]

- c) Along what other angles θ (besides $\theta = 0$) will “petal in petal” also occur?

List all such angles θ in radians. $\frac{2\pi}{3}$, $\frac{4\pi}{3}$ [2pts]

- d) For what angles $0 \leq \theta \leq \frac{\pi}{2}$ will this curve intersect the circle $r = 2$? Show all the work that leads to your answer(s). [3pts]

a) when $\theta = 0$
 $2 + 6 = 8$

$2 + 6\cos 3(\theta + \pi) = -r$
 $-2 = 2 + 6\cos 3\theta$
 $6\cos 3\theta = -4$
 $\cos 3\theta = -\frac{2}{3}$
 $3\theta = \cos^{-1}(-\frac{2}{3})$
 $\theta = \frac{1}{3}\cos^{-1}(-\frac{2}{3})$
 $\theta = \frac{1}{3}\cos^{-1}(-\frac{2}{3}), \frac{\pi}{6}, \frac{\pi}{2}$

$2 = 2 + 6\cos 3\theta$
 $6\cos 3\theta = 0$
 $\cos 3\theta = 0$
 $3\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$
 $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}$

$0 \leq \theta \leq \frac{\pi}{2}$
 12 intersections

$$9 \text{ petals} \quad \frac{2\pi}{11} \times 8 = \frac{16\pi}{11} \quad \frac{2\pi}{11} \times 9 = \frac{18\pi}{11} - \frac{3\pi}{2} = \frac{36\pi}{22} - \frac{33\pi}{22} = \frac{3\pi}{22}$$

$$\frac{16\pi}{11} \times \frac{1}{2} = \frac{8\pi}{11} \quad \frac{18\pi}{11} \times \frac{1}{2} = \frac{9\pi}{11} \quad \frac{3\pi}{22} - \frac{33\pi}{22} = -$$

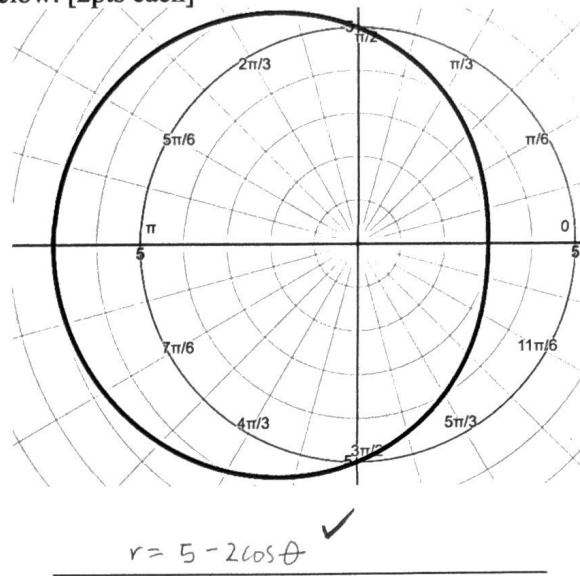
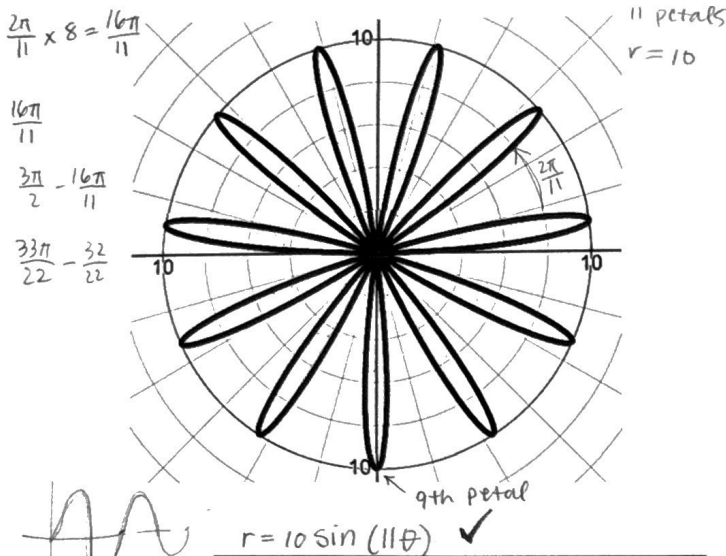
cos

$$x = -2$$

$$5 + x = 3$$

$$5 - x = 7$$

5. a) Write polar equations that match the two graphs below. [2pts each]



$$r = \frac{12}{\pi}$$

$$22 = x\pi$$

Answer the following questions about the polar rose graphed above on the left.

b) How many radians will it take to create a single petal? $\frac{\pi}{11}$ [1pt]

c) Give a polar equation of the first line of symmetry in the first quadrant. $\frac{\pi}{22}$ [1pt]

$$\frac{2\pi}{22} = 11$$

$$11p = 2\pi$$

$$p = \frac{2\pi}{11}$$

6. Derive a general polar equation of a **horizontal** line that passes through the polar point (k, α) (where both k and α are constants). Give your answer in terms of k , α and θ . [2pts]

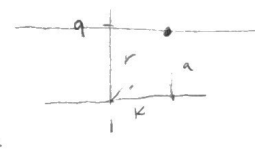
$$y = k \sin \theta$$

$$k = a \cos \theta$$

$$a = \frac{k}{\cos \theta} = k \sec \theta$$

$$k = a \sin \theta$$

$$a = k \csc \theta$$



$$r^2 = a^2 + k^2$$

$$\frac{a}{b} = \frac{c}{d}$$

-2

$$r = k \csc \theta$$

7. Convert the equation: $r = \frac{3}{1 - \sin \theta}$ into rectangular form. Rewrite your answer (without a square root sign) to simplify it into a form of a common function. What is the shape of the graph? [4pts]

$$r = \frac{3}{1 - \sin \theta}$$

$$r(1 - \sin \theta) = 3$$

$$r - r \sin \theta = 3$$

$$r = r \sin \theta + 3$$

$$r^2 = (r \sin \theta + 3)^2$$

$$x^2 + y^2 = (y + 3)^2$$

$$x^2 + y^2 = y^2 + 6y + 9$$

$$x^2 = 6y + 9$$

$$6y = x^2 - 9$$

$$y = \frac{x^2 - 9}{6} = \frac{(x+3)(x-3)}{6}$$

Rectangular form: $y = \frac{x^2 - 9}{6}$. What shape is the graph? parabola

x	y
1	$-\frac{4}{3}$
2	$-\frac{5}{6}$
3	0
4	$\frac{7}{6}$
5	$\frac{8}{3}$

